Two Clocks: the ± 1 Meters, Depth Supply, and Unconditional Escape Bounds

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Abstract

We complete the run–budget clock of the companion note: the b=1 runs of a Syracuse valuation word meter 2-adic approach to the ghost -1 (depth decrement exactly 1 per step), and — new here — the b=2 runs meter approach to the ghost $+1$, the *trivial cycle*, with depth decrement exactly 2 per step and a parity handoff. Every letter is a depth reading against one of the three rational points -1, $+1$, $-\frac{1}{3}$, and the word parses uniquely into alternating meter readings (machine-verified). Ascent decomposes exactly into a two-clock budget. Acyclicity makes every reading consume a congruence point *without replacement*, giving unconditional results: a divergent orbit upcrosses the level 2^k at most 2^{k-3} times; an acyclic orbit dwells in a band of ratio ≤ 2 at most $O(\log Y)$ consecutive steps (the $D = \frac{1}{2}$ strip carries exactly *one* word per phase); hence a divergent orbit spends at most $O(X \log X)$ steps of its entire lifetime below height X. Target 2 is then isolated as a *joint* calibration conjecture for two scalar channels — the -1 depth law and the $+1$ *parity* law — and we show the first channel alone cannot suffice: a natural-depth orbit can still ascend through pure parity bias, along the $(1, 1, 2)$-channel whose limit is the negative rational cycle $-19/11$.

1 The trichotomy and the two clocks

Throughout $S(x) = (3x + 1)/2^b$, $b = v_2(3x + 1)$, acting on odd integers.

Lemma 1 (Letter trichotomy). *For odd x, exactly one of the following holds:*

(i) $x \equiv 3 \pmod 4$*, i.e.* $v_2(x + 1) \ge 2$*; then* $b = 1$*;*

(ii) $x \equiv 1 \pmod 8$*, i.e.* $v_2(x - 1) \ge 3$*; then* $b = 2$*;*

(iii) $x \equiv 5 \pmod 8$*, i.e.* $v_2(x - 1) = 2$*; then* $b \ge 3$*.*

Uniformly $b = v_2\big(x + \frac{1}{3}\big)$ *in* $\mathbb{Z}_2$*: every letter is a 2-adic depth reading against one of* -1, $+1$, $-\frac{1}{3}$.

Proof. Write $3x + 1 = 3(x + 1) - 2 = 3(x - 1) + 4$. If $v_2(x + 1) \ge 2$ then $v_2(3(x + 1)) \ge 2$ and subtracting 2 gives $v_2 = 1$: $b = 1$. If $x - 1 = 2^h u$, u odd, $h \ge 3$: $3x + 1 = 4(3 \cdot 2^{h-2}u + 1)$ with the inner factor odd, so $b = 2$. If $h = 2$: $3x + 1 = 4(3u + 1)$ with $3u + 1$ even, so $b \ge 3$. Finally $v_2(3x + 1) = v_2(3) + v_2(x + \frac{1}{3}) = v_2(x + \frac{1}{3})$. □

Lemma 2 (-1 clock; companion note)**.** *If* $g = v_2(x + 1) \ge 2$ *then* $S(x) + 1 = 3(x + 1)/2$*: the depth decreases by exactly 1 per step, the maximal b=1 run from x has length exactly* $g - 1$*, and* $g \le \log_2(x + 1)$ *for* $x > 0$*.*

Lemma 3 (+1 clock)**.** *If $h = v_2(x-1) \geq 3$ then $b = 2$ and*

$$S(x) - 1 = \frac{3(x-1)}{4} :$$

the depth against $+1$ *decreases by exactly* 2 *per step. The maximal* b=2 *run from* x *has length exactly* $\lfloor (h-1)/2 \rfloor$*, with terminal depth* 1 *if* h *is odd — the run hands off directly to a* b=1 *run — and terminal depth* 2 *if* h *is even — the next letter is* $b \geq 3$*. Moreover* $h \leq \log_2(x-1)$ *for* $x > 1$*, and* b=2 *forever* $\iff$ $x = 1$*:* the +1 meter measures 2-adic proximity to the trivial cycle.

Proof. $S(x) = (3x+1)/4$ by Lemma 1, and $S(x)-1 = (3x-3)/4 = 3(x-1)/4$, so $v_2(S(x)-1) = h-2$. The run continues while the current depth is ≥ 3, giving length $\lfloor (h-1)/2 \rfloor$ and terminal depth $h - 2\lfloor (h-1)/2 \rfloor \in \{1, 2\}$ according to parity; terminal depth 1 means $x \equiv 3 \pmod 4$ (case (i)), terminal depth 2 is case (iii). Positivity gives the cap; $x = 1$ is the fixed point of $x \mapsto (3x+1)/4$. $\square$

Proposition 1 (Alternating parse)**.** *Every Syracuse word is reproduced exactly by reading only the meters: emit* $g-1$ *ones and jump* $x \mapsto 3^{g-1}(x+1)/2^{g-1} - 1$*; emit* $\lfloor (h-1)/2 \rfloor$ *twos and jump* $x \mapsto 3^{\lfloor (h-1)/2 \rfloor}(x-1)/4^{\lfloor (h-1)/2 \rfloor} + 1$*; at depth-2 points emit the single letter* $b = v_2(x + \frac{1}{3}) \geq 3$*. (Machine-verified on* 2,000 *random trajectories of length* 300*; clock laws verified exactly on* 20,000 *samples.)*

Proposition 2 (Two-clock budget identity)**.** *Over any orbit window of* N *letters, writing* g_i *for the* 1*-run depths,* h_j *for the* 2*-run depths and* $b_\ell \geq 3$ *for the big letters,*

$$\log_2 \frac{x_N}{x_0} = (\log_2 3 - 1)\sum_i (g_i - 1) - (2 - \log_2 3)\sum_j \left\lfloor \frac{h_j - 1}{2} \right\rfloor - \sum_\ell (b_\ell - \log_2 3) + E,$$

with $0 \leq E \leq N/(3x_{\min} \ln 2)$*. Ascent is therefore financed by exactly three channels: deep* -1 *readings (*$+0.585$ *per unit run length), shallow-and-odd* $+1$ *readings (*-0.415 *per* 2*-letter), and avoidance of big letters (*$-(b - 1.585)$ *each).*

2 Depth supply: readings are without replacement

Lemma 4 (Depth supply)**.** *Let the forward orbit be acyclic (e.g. divergent). For* $2 \leq g \leq k+1$*, the number of times the orbit* ever *visits a point* $x \in [2^k, 2^{k+1})$ *with* $v_2(x+1) \geq g$ *is at most* $\max(2^{k-g}, 1)$*, and likewise for* $v_2(x-1) \geq g$*. In particular each* b=1 *step permanently consumes one point of the* $\equiv 3 \pmod 4$ *pool of its dyadic band, and the unique deepest* -1 *point of band* k *is the truncated ghost* $2^{k+1} - 1$.

Proof. An acyclic orbit visits each integer at most once; the points of depth $\geq g$ in the band form the residue class $-1 \bmod 2^g$ (resp. $+1$), of cardinality 2^{k-g} for $g \leq k$ and 1 for $g = k+1$. $\square$

Theorem 1 (Upcrossing bound)**.** *A divergent orbit makes at most* 2^{k-3} *upward crossings of the level* 2^k*, ever.*

Proof. Only b=1 steps ascend, and they multiply by less than 2; so the step crossing 2^k starts at some $x \in [2^{k-1}, 2^k)$ with $x \equiv 3 \pmod 4$. By acyclicity these starting points are pairwise distinct, and the pool has 2^{k-3} elements. (Using $x \geq (2^{k+1}-1)/3$ refines the constant to $2^k/12 + O(1)$.) $\square$

3 Logarithmic dwell and the escape rate

Lemma 5 (The dyadic strip carries one word)**.** *For $D = \frac{1}{2}$ the confinement window $[j \log_2 3 - D, j \log_2 3 + D]$ has length 1 and contains exactly one integer for every j (irrationality of $\log_2 3$); hence the exact count of the parent note satisfies $N_{1/2}(p) = 1$ for all p: the unique confined word is the Sturmian word of slope $\log_2 3$. (Verified by the exact big-integer recursion to $p = 1600$.) Allowing an arbitrary phase, the number of distinct length-p blocks appearing in any width-1-confined word is at most $p + 1$, by the three-distance argument.*

Theorem 2 (Logarithmic dwell at ratio ≤ 2)**.** *An acyclic positive orbit remains within a band $[Y, WY]$, $W \leq 2$, for at most $3 \log_2 Y + O(1)$ consecutive steps.*

Proof. Set $p = \lceil \log_2 Y \rceil$. The segment's word is confined to a strip of width $1 + O(M/Y)$, so by Lemma 5 it has at most $2(p + 1)$ distinct length-p blocks (the $O(M/Y)$ widening can double the integer choice at no more than one position once $Y \gg M$). A segment of length $M > 2(p + 1) + p$ contains two positions carrying the same length-p block; both lie in the band, so they differ by less than $Y \leq 2^p < 2^{p+1}$, and repetition rigidity (parent program, Lean-verified) forces the two points to be *equal* — an integer cycle, contradicting acyclicity. Hence $M \leq 3p + O(1)$. ☐

Theorem 3 (Escape rate)**.** *A divergent orbit spends at most $O(X \log X)$ steps of its entire lifetime at heights $\leq X$.*

Proof. Visits to the dyadic band $[2^k, 2^{k+1})$ are bounded by upcrossings of 2^k plus downcrossings of 2^{k+1} plus one; downcrossings of a level exceed its upcrossings by at most one, so by Theorem 1 the band is visited at most $2^{k-3} + 2^{k-2} + 2 \leq 2^{k-1}$ times. Each visit lasts at most $3k + O(1)$ steps by Theorem 2. Summing $\sum_{k \leq \log_2 X} 2^{k-1}(3k + O(1)) = O(X \log X)$. ☐

Remark. *Theorem 3 says divergence, if it occurs, is* brisk*: the orbit cannot oscillate indefinitely through bounded regions; its occupation time below every height is finite and near-linear. All ingredients are elementary given repetition rigidity.*

4 Target 2 isolated: the joint calibration conjecture

Under the natural (2-adic Haar) law the meters read $\mathbb{P}[g = j] = 2^{-(j-1)}$ ($j \geq 2$, mean 3), $\mathbb{P}[h = j] = 2^{-(j-1)}$ ($j \geq 2$, mean 3, $\mathbb{P}[h \text{ odd}] = \frac{1}{3}$), and the budget of Proposition 2 gives the familiar drift $\log_2 3 - 2 < 0$: descent. Measured on random trajectories (windows of 60 letters; ascending = end $\geq 4\times$ start, never below start):

channel	natural	typical windows	ascending windows
-1 depth mean $\mathbb{E}[g]$	3.000	3.053	3.990
$+1$ depth mean $\mathbb{E}[h]$	3.000	3.020	3.196
$+1$ parity $\mathbb{P}[h \text{ odd}]$	0.333	0.358	**0.630**
big letters per step	0.250	0.235	0.093

The exact budget decomposition of the ascending windows (+23.82 from 1-runs, −5.69 from 2-letters, −11.17 from big letters = +6.95, matching the measured net exactly) shows ascent drawing on *both* clocks: deeper −1 readings *and* an odd-parity bias of the +1 readings, the latter being precisely avoidance of big letters after 2-runs.

Conjecture 1 (Two-clock calibration; Target 2)**.** *Along any divergent positive orbit the empirical meter laws calibrate to natural: the run-start depth law tends to* $2^{-(g-1)}$ and *the handoff parity tends to* $\frac{1}{3}$*. (Either statement is to be read in Cesàro form; by Proposition 2 joint calibration forces mean drift* $\log_2 3 - 2 < 0$*, contradicting divergence.)*

Remark (The g-law alone cannot suffice)**.** *A counterexample cannot be excluded by the* -1 *channel only: an orbit with perfectly natural depth law (*$\mathbb{E}[g] = 3$*) but pure parity bias (*$h \equiv 3$ *always) follows the letter pattern* $(1, 1, 2)^\infty$*, which has mean* $\frac{4}{3} < \log_2 3$ *and ascends. This channel's limit object is the* negative rational cycle $x = 19/(2^4 - 3^3) = -19/11$*, a ghost: the conjecture's parity half is exactly the statement that positive integers cannot ride the* $-19/11$*-ghost forever. Any proof of Target 2 must therefore meter both clocks.*

Remark (Bridge to the mixed $(2, 3)$-adic tower)**.** *After a* 1*-run of length* $\ell = g - 1$ *from* x *with* $x + 1 = 2^g m$ *(*m *odd), the next handoff reading is*

$$h \;=\; 1 + v_2\big(3^\ell m - 1\big) :$$

the $+1$ *meter literally reads the* 2*-adic valuation of* $3^\ell m - 1$*. Its parity — the channel ascending orbits bias — is a joint condition on powers of* 3 *against the* higher *bits* m *of the current point, the lifting-the-exponent regime (for the truncated ghosts* $m = 1$ *the handoffs are fully deterministic:* $v_2(3^\ell - 1) = 1$ *for* ℓ *odd,* $= 2 + v_2(\ell)$ *for* ℓ *even). This places the band-crossing certificate of Target 2 squarely in the joint* $2^a 3^b$ *tower of the parent program: the certificate must meter* $v_2(3^\ell m - 1)$*, not* m *alone.*

Remark (What remains, honestly)**.** *The supply bounds of Lemma 4 are permanent but the pools regenerate with height: a divergent orbit gains fresh deep points as it climbs, so counting alone cannot close Conjecture 1. What the two-clock structure changes is the* shape *of the wall: quenched normality of the full word is no longer needed — only two scalar renewal channels, each a congruence reading at handoff points, with the* $(2, 3)$*-tower coupling above as the natural attack surface.*